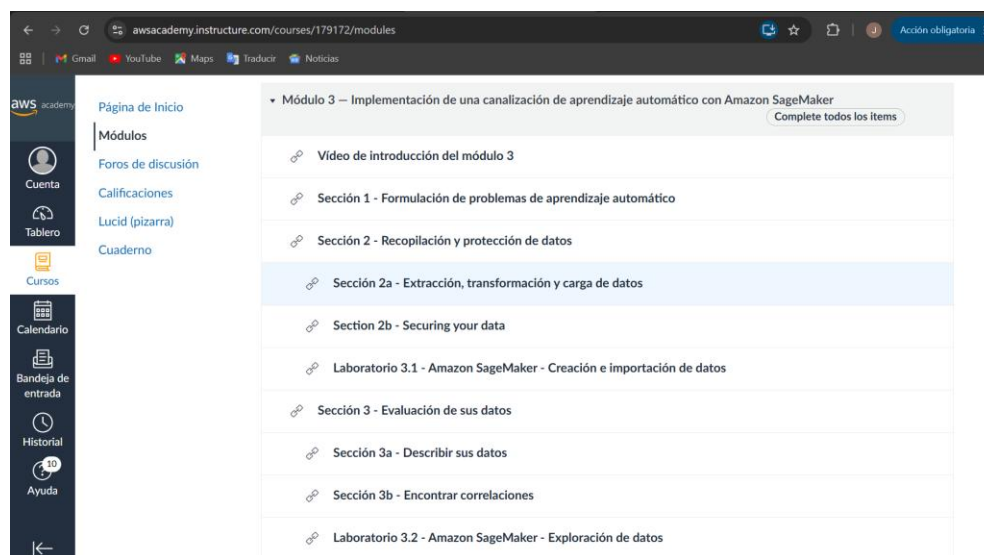
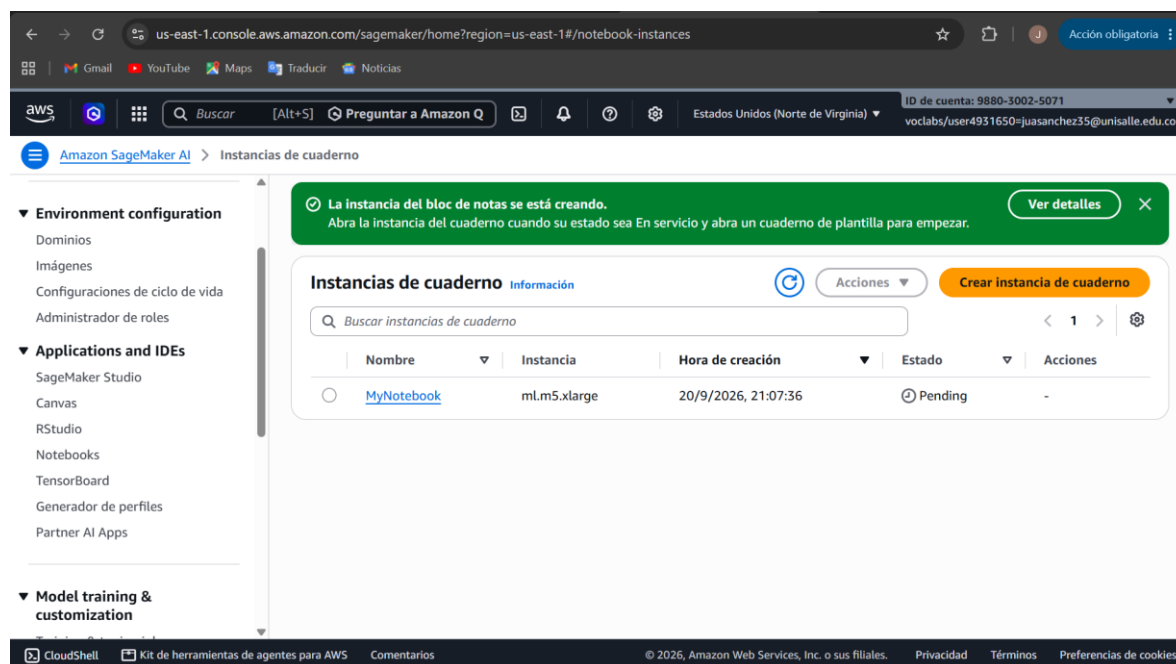


Evidencias

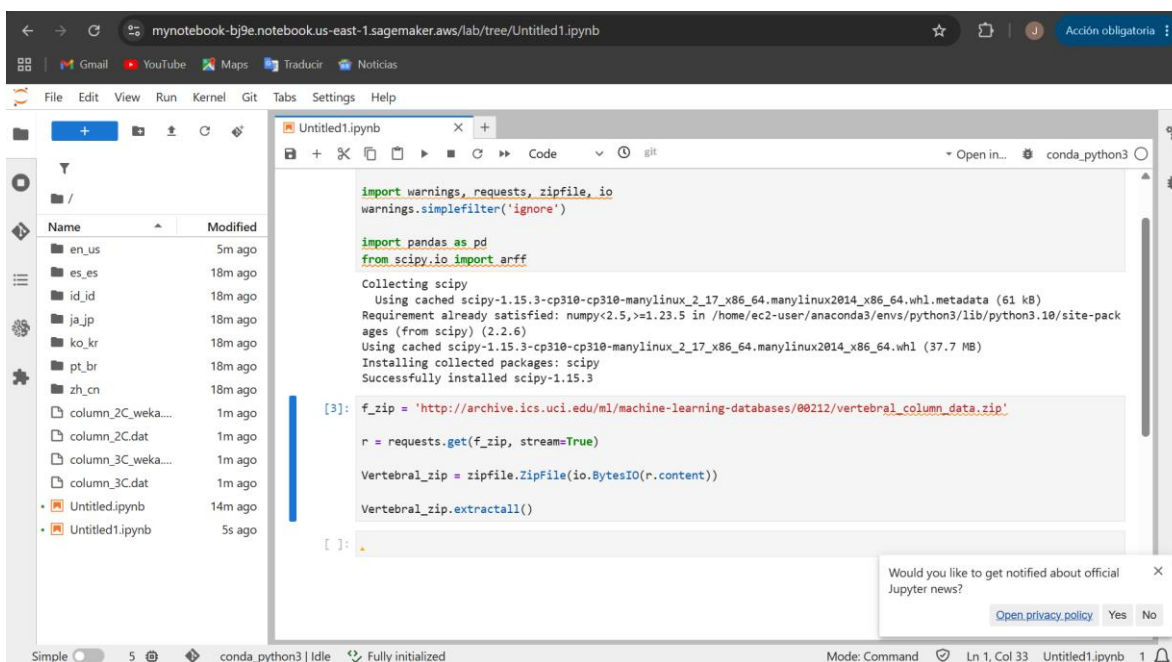
Captura de acceso a la plataforma



Instancia de cuaderno creada



Archivos en el explorador de jupyterlab



The screenshot shows the JupyterLab interface. On the left, the file explorer displays a directory structure with files like `en_us`, `es_es`, `id_id`, `ja_jp`, `ko_kr`, `pt_br`, `zh_cn`, `column_2C.weka...`, `column_2C.dat`, `column_3C.weka...`, `column_3C.dat`, `Untitled.ipynb`, and `Untitled1.ipynb`. The main area shows the code editor for `Untitled1.ipynb`. The code includes imports for `warnings`, `requests`, `zipfile`, `io`, `pandas`, and `arff`. It also shows the installation of `scipy` and the download of a dataset from a URL. The output of the code execution is visible, showing the installation of `scipy` and the download of the dataset.

```
import warnings, requests, zipfile, io
warnings.simplefilter('ignore')

import pandas as pd
from scipy.io import arff

Collecting scipy
Using cached scipy-1.15.3-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (61 kB)
Requirement already satisfied: numpy<2.5, >=1.23.5 in /home/ec2-user/anaconda3/envs/python3/lib/python3.10/site-pack
ages (from scipy) (2.2.6)
Using cached scipy-1.15.3-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (37.7 MB)
Installing collected packages: scipy
Successfully installed scipy-1.15.3

[3]: f_zip = 'http://archive.ics.uci.edu/ml/machine-learning-databases/00212/vertebral_column_data.zip'

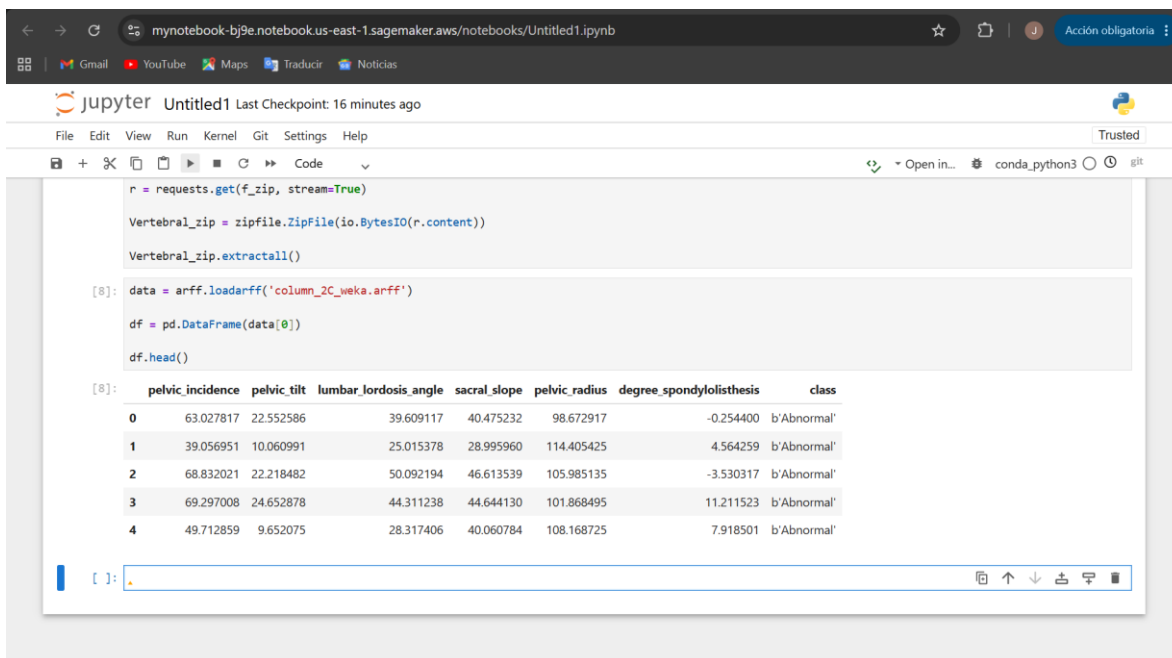
r = requests.get(f_zip, stream=True)

Vertebral_zip = zipfile.ZipFile(io.BytesIO(r.content))

Vertebral_zip.extractall()

[ ]:
```

Cargar dataset en Pandas



The screenshot shows the JupyterLab interface with the code editor for `Untitled1.ipynb`. The code includes the same imports as before, plus `arff` and `pandas`. It shows the download of a dataset from a URL and the loading of the dataset into a `pandas` `DataFrame`. The output of the code execution is visible, showing the `DataFrame` structure and the first few rows of data.

```
r = requests.get(f_zip, stream=True)

Vertebral_zip = zipfile.ZipFile(io.BytesIO(r.content))

Vertebral_zip.extractall()

[0]: data = arff.loadarff('column_2C.weka.arff')

df = pd.DataFrame(data[0])

df.head()

[0]:
```

	pelvic_incidence	pelvic_tilt	lumbar_lordosis_angle	sacral_slope	pelvic_radius	degree_spondylolisthesis	class
0	63.027817	22.552586	39.609117	40.475232	98.672917	-0.254400	b'Abnormal'
1	39.056951	10.060991	25.015378	28.995960	114.405425	4.564259	b'Abnormal'
2	68.832021	22.218482	50.092194	46.613539	105.985135	-3.530317	b'Abnormal'
3	69.297008	24.652878	44.311238	44.644130	101.868495	11.211523	b'Abnormal'
4	49.712859	9.652075	28.317406	40.060784	108.168725	7.918501	b'Abnormal'

Laboratorio 3.2

Acceso a Mynotebook

The screenshot shows the Mynotebook interface in a web browser. The URL is `mynotebook-hdmt.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/3_2-machinelearning.ipynb`. The interface includes a file explorer on the left showing the `en_us` directory with a file named `3_2-machinelearning.ipynb` modified 11m ago. The main content area displays the 'Lab 3.2 - Student Notebook' overview, which includes an introduction to the business scenario and information about the dataset. A notification pop-up asks if the user wants to get notified about official Jupyter news.

Lab 3.2 - Student Notebook

Overview

This lab is a continuation of the guided labs in Module 3.

Introducing the business scenario

You work for a healthcare provider, and want to improve the detection of abnormalities in orthopedic patients.

You are tasked with solving this problem by using machine learning (ML). You have access to a dataset that contains six biomechanical features and a target of *normal* or *abnormal*. You can use this dataset to train an ML model to predict if a patient will have an abnormality.

About this dataset

This biomedical dataset was built by Dr. Henrique da Mota during a medical residence period at the Centre Médico-Chirurgical de Réadaptation des Massues, Lyon.

The first task consists in classifying patients as belonging to one of three categories:

Captura laboratorio 3.2 finalizado

The screenshot shows the Mynotebook interface with the lab completed. The left pane displays a heatmap of the correlation matrix for the dataset features. The right pane shows the code used to generate the heatmap and a congratulatory message.

Heatmap Data

	pelvic_incidence	pelvic_tilt	lumbar_flexion_angle	sacral_slope	pelvic_radius	degree_spondylosiathesis	class
pelvic_incidence	1.00	0.63	0.72	0.81	-0.25	0.84	0.35
pelvic_tilt	0.63	1.00	0.43	0.06	0.03	0.40	0.33
lumbar_flexion_angle	0.72	0.43	1.00	0.60	-0.08	0.53	0.31
sacral_slope	0.81	0.06	0.60	1.00	-0.34	0.52	0.21
pelvic_radius	-0.25	0.03	-0.08	-0.34	1.00	-0.03	-0.31
degree_spondylosiathesis	0.84	0.40	0.53	0.52	-0.03	1.00	0.44
class	0.35	0.33	0.31	0.21	-0.31	0.44	1.00

```
[ ]: sns.heatmap(corr_matrix, cmap=colormap, annot=True, fmt=".2f")
#ax.set_yticklabels(column_names);
plt.show()
```

```
[ ]: pd.plotting.scatter_matrix(df, figsize=(12,12))
plt.show()
```

By using `seaborn`, you can visualize the correlation as a *heatmap*.

```
[ ]: import seaborn as sns
# Plot figsize
fig, ax = plt.subplots(figsize=(10, 10))
# Generate Color Map
# colormap = sns.diverging_palette(220, 10, as_cmap=True)
colormap = sns.color_palette("BrBG", 10)
# Generate Heat Map, allow annotations and place floats in map
sns.heatmap(corr_matrix, cmap=colormap, annot=True, fmt=".2f")
#ax.set_yticklabels(column_names);
plt.show()
```

Challenge task: Find other data from the UCI Machine Learning Repository. Using the previous code for reference, go explore!

Congratulations!

You have completed this lab, and you can now end the lab by following the lab guide instructions.

Laboratorio 3.3

The screenshot shows a Jupyter Notebook titled "3_3-machinelearning.ipynb" in a web browser. The notebook contains three code cells:

```
[21]: df_car['aspiration'].value_counts()

[21]: aspiration
      std      168
      turbo      37
      Name: count, dtype: int64

[22]: df_car = pd.get_dummies(df_car, columns=['aspiration'], drop_first=True)

[23]: df_car.head()
```

The output of the third cell is a table with 10 columns: num-of-doors, num-of-cylinders, doors, cylinders, drive-wheels_4wd, drive-wheels_fwd, drive-wheels_rwd, aspiration_turbo, and aspiration_std. The table shows the first five rows of data.

	num-of-doors	num-of-cylinders	doors	cylinders	drive-wheels_4wd	drive-wheels_fwd	drive-wheels_rwd	aspiration_turbo	aspiration_std
0	two	four	2.0	4	False	False	True	False	168
1	two	four	2.0	4	False	False	True	False	37
2	two	six	2.0	6	False	False	True	False	
3	four	four	4.0	4	False	True	False	False	
4	four	five	4.0	5	True	False	False	False	

Challenge task: Go back to the beginning of this lab, and add other columns to the dataset. How would you encode the values of each column? Update the code to include some of the other features.